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Soc analyst

Lab environment

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Executive Summary

This project presents a Security Operations Centre (SOC) laboratory environment designed to simulate real-world cybersecurity monitoring using a combination of attack simulation and log analysis tools.

The environment was built using DVWA (Damn Vulnerable Web Application), Splunk SIEM, Apache web server logs, and Kali Linux as the attack platform. Controlled attacks such as brute force login attempts and web enumeration were simulated using Hydra and manual requests to generate realistic security events.

The primary focus of the project was to observe how malicious activity appears in logs and how a SIEM system can be used to analyses and visualize this behaviour. The outcome of the project demonstrates practical understanding of security monitoring, log analysis, and attack behaviour detection in a controlled environment.

Introduction

The purpose of this SOC laboratory project is to simulate a real-world cybersecurity monitoring environment where security events can be generated, collected, and analysed.

Cybersecurity monitoring is essential in modern IT environments because organizations must continuously detect and respond to malicious activity such as unauthorized access attempts, scanning behaviour, and system exploitation attempts.

The aim of this project was to build a functional SOC environment that allows for:

- Simulating cyber-attacks in a controlled lab
- Collecting and analysing system and web logs
- Understanding how attacker behaviour appears in SIEM tools
- Gaining practical experience with security monitoring technologies

This project helps bridge the gap between theoretical cybersecurity knowledge and real-world SOC operations.

Tools and Technologies

- Splunk SIEM was configured locally on 127.0.0.1 and used to ingest and analyse security logs from the SOC lab environment. The primary data source was /var/log/apache2/access.log, which captured web server activity including login attempts and HTTP requests.

  Logs generated from the Kali Linux machine and DVWA attack simulations were successfully integrated into Splunk, enabling centralized monitoring of authentication attempts, web traffic, and system events.

This setup allowed real-time log ingestion, search queries, and visualization of security events, supporting effective analysis of attack patterns within the SOC environment.

- DVWA (Damn Vulnerable Web Application)
- Kali Linux
- Hydra (password attack tool)
- Apache Web Server

Attack simulation

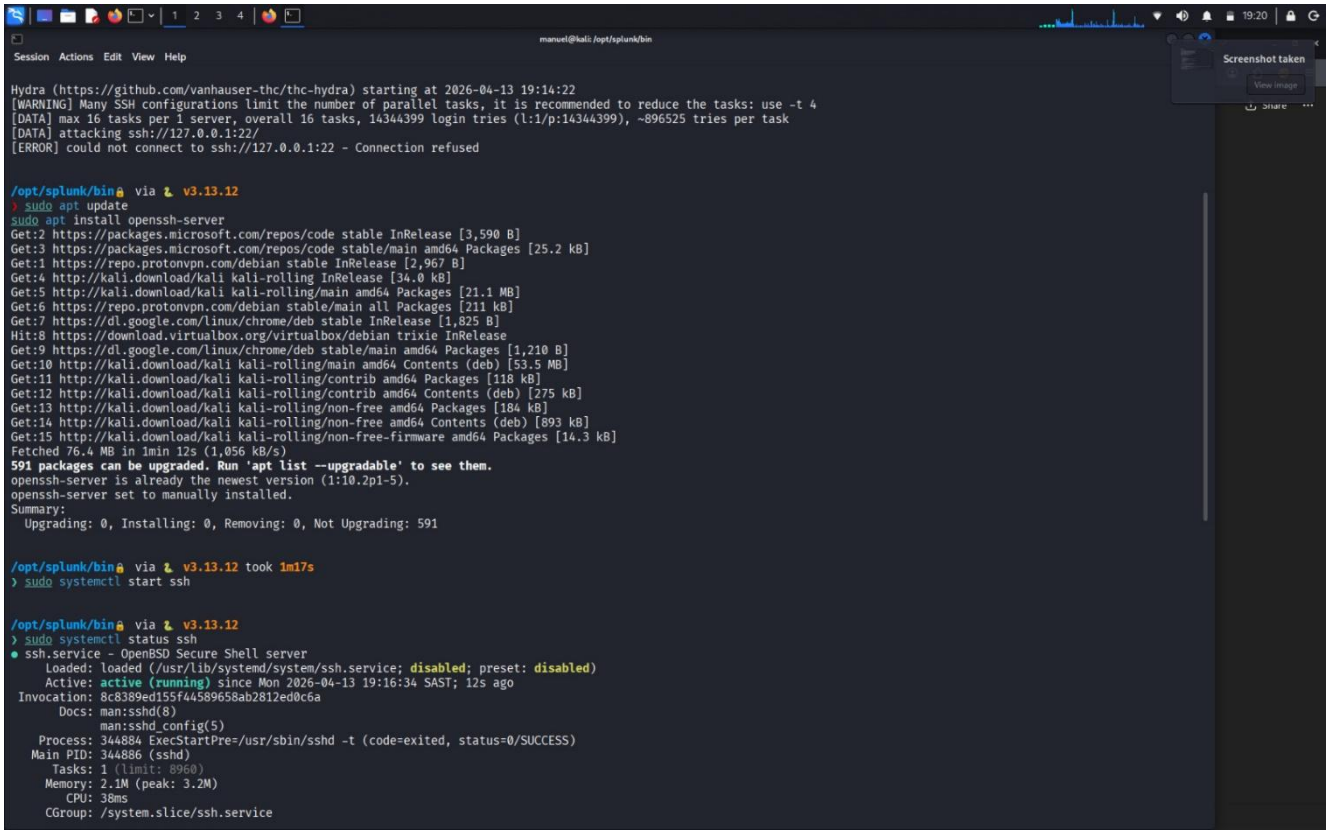
SSH was configured in the SOC lab environment to enable secure remote access and controlled interaction between systems within the network.

The main purpose of adding SSH in this project was to simulate real-world administrative access in enterprise environments, where security analysts and system administrators remotely manage servers and endpoints.

In this SOC setup, SSH served the following functions:

- Provided secure remote login to the Kali Linux and target systems
- Allowed execution of commands for monitoring and system administration
- Generated authentication logs that were ingested into Splunk for analysis
- Enabled simulation of unauthorized access attempts (e.g., brute force testing) to evaluate detection capabilities

By including SSH in the environment, the project was able to replicate realistic login activity patterns, which are essential for SOC monitoring, alerting, and investigating potential credential-based attacks.



Above the screenshot shows that the ssh is running on the machine.

BRUTEFORCE

A controlled brute force simulation was executed in a local lab environment against the DVWA login page. The purpose of the test was to generate authentication failure events and analyze how repeated login attempts appear in web server logs and SIEM systems.

Observations

- Multiple automated login attempts were generated against the /DVWA/login.php endpoint
- The web server recorded repeated HTTP POST requests
- Authentication failures were consistently logged in Apache access logs
- The activity pattern showed high-frequency login attempts from a single source

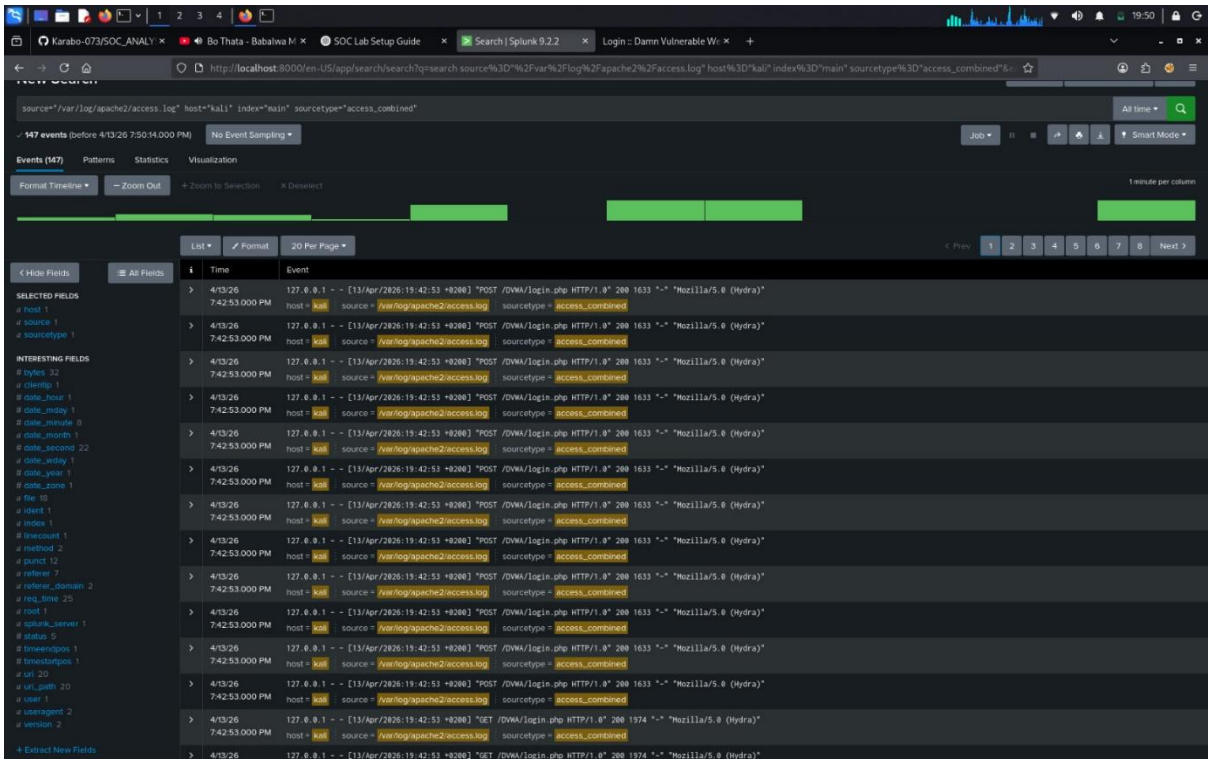
```
~ via 🐍 v3.13.12
❯ hydra -l admin -P small.txt 127.0.0.1 http-post-form "/DVWA/login.php:username=^USER^&password=^PASS^:Login failed"
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these
*** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-13 19:37:08
[DATA] max 16 tasks per 1 server, overall 16 tasks, 50 login tries (l:1/p:50), ~4 tries per task
[DATA] attacking http-post-form://127.0.0.1:80/DVWA/login.php:username=^USER^&password=^PASS^:Login failed
[80][http-post-form] host: 127.0.0.1   login: admin   password: 123456789
[80][http-post-form] host: 127.0.0.1   login: admin   password: 123456
[80][http-post-form] host: 127.0.0.1   login: admin   password: 12345
[80][http-post-form] host: 127.0.0.1   login: admin   password: password
[80][http-post-form] host: 127.0.0.1   login: admin   password: iloveyou
[80][http-post-form] host: 127.0.0.1   login: admin   password: princess
[80][http-post-form] host: 127.0.0.1   login: admin   password: 1234567
[80][http-post-form] host: 127.0.0.1   login: admin   password: nicole
[80][http-post-form] host: 127.0.0.1   login: admin   password: monkey
[80][http-post-form] host: 127.0.0.1   login: admin   password: rockyou
[80][http-post-form] host: 127.0.0.1   login: admin   password: 12345678
[80][http-post-form] host: 127.0.0.1   login: admin   password: abc123
[80][http-post-form] host: 127.0.0.1   login: admin   password: daniel
[80][http-post-form] host: 127.0.0.1   login: admin   password: babygirl
[80][http-post-form] host: 127.0.0.1   login: admin   password: lovely
[80][http-post-form] host: 127.0.0.1   login: admin   password: jessica
1 of 1 target successfully completed, 16 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-13 19:37:09
```

The above screenshot shows that 20 passwords were discovered using hydra.

THE RESULTS ON SPLUNK

On Splunk I ran index=main "POST /DVWA/login.php"to show multiple attempts. the below screenshot shows that.



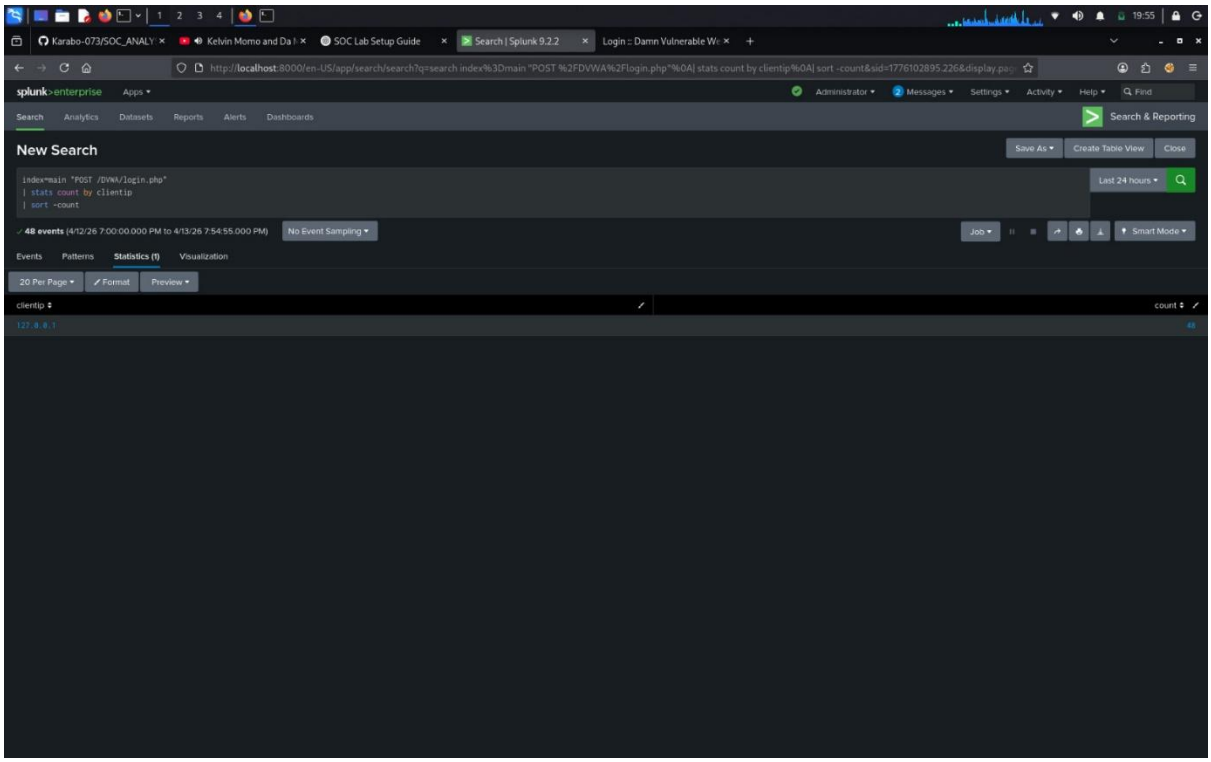
Searching

index=main "POST /DVWA/login.php"

| stats count by clientip

| sort -count

on Splunk it showed me which IP is attacking most.



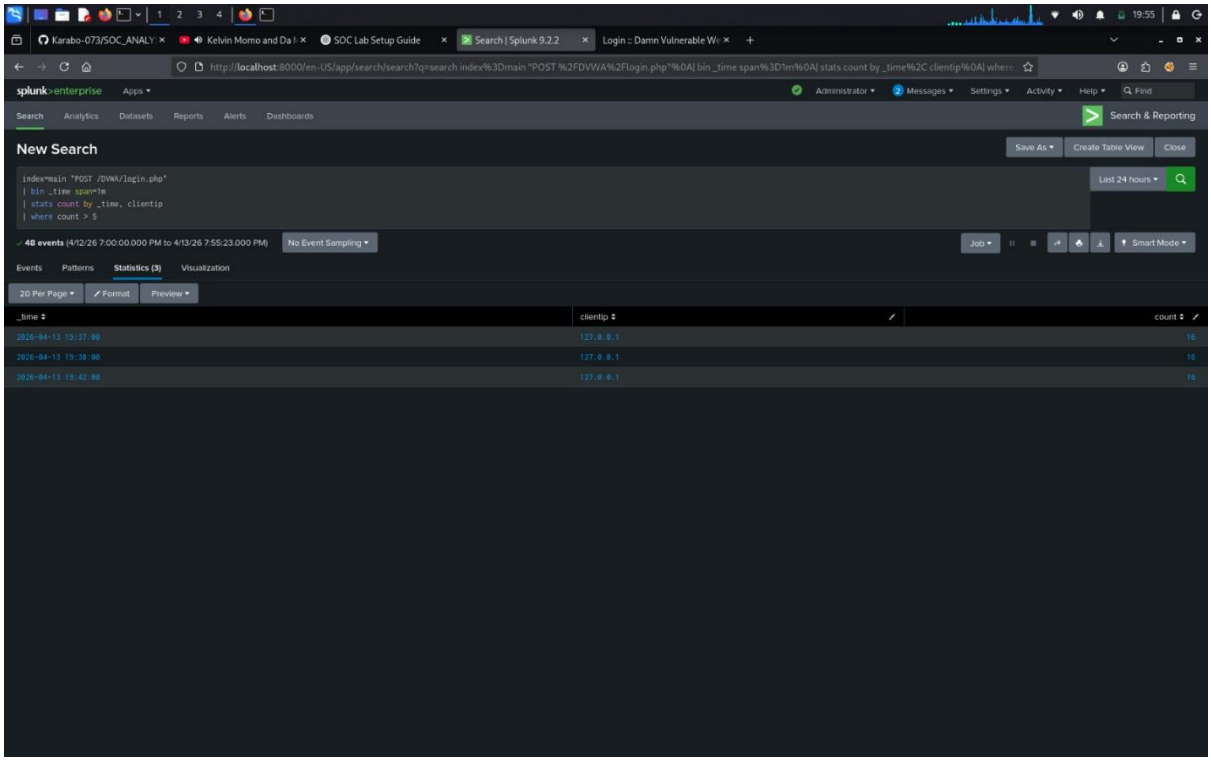
index=main "POST /DVWA/login.php"

| bin _time span=1m

| stats count by _time, clientip

| where count > 5

This means more than 5 login attempts per minute = brute force



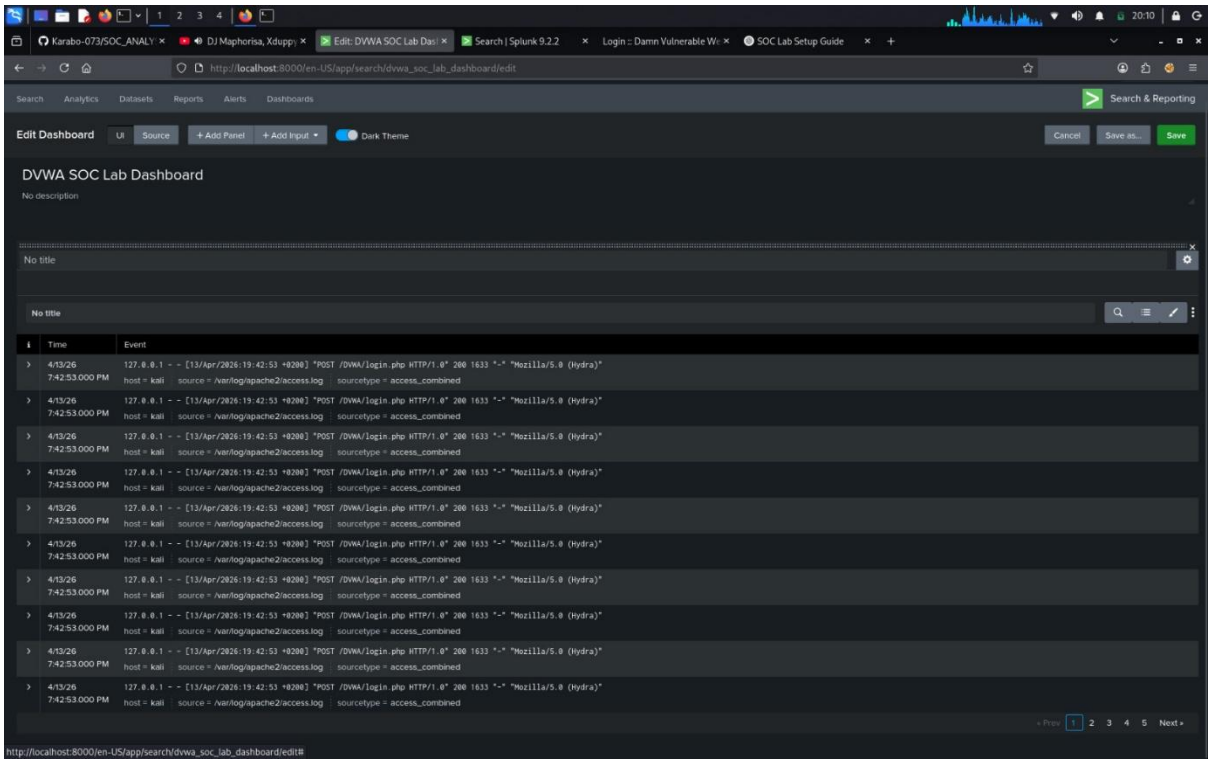
Building a soc dashboard

Add a panel

1. Attack volume

index=main "POST /DVWA/login.php"

| timechart count



2. Top attacker Ips

3. Login attempts over time

Built a SOC lab using Splunk to ingest Apache web logs from a DVWA environment. Simulated brute-force attacks using Hydra and developed detection queries and alerts to identify high-frequency login attempts.

Summary

This SOC project demonstrates the design and implementation of a local Security Operations Centre (SOC) laboratory using DVWA, Splunk SIEM, Apache web server logs, Kali Linux, Hydra, and SSH.

The environment was built to simulate real-world cyber attacks such as brute force login attempts and web enumeration. These attacks generated authentication and web access logs, which were ingested into Splunk for centralized monitoring and analysis.

Splunk was configured locally on 127.0.0.1 to collect logs from /var/log/apache2/access.log, enabling real-time visibility of login attempts and HTTP traffic. SSH was also configured to simulate administrative access and generate authentication logs for monitoring.

Brute force activity was simulated against the DVWA login page, producing repeated failed login attempts that were successfully identified in Splunk using search queries and time-based analysis. Dashboards were created to visualise attack volume, top source IPs, and login attempt frequency over time.

Overall, the project demonstrates practical SOC skills including log analysis, attack simulation, SIEM usage, dashboard creation, and basic detection of brute force activity in a controlled environment.